Unit no. ° 9

PROPORTIONS AND CHANGES

21-29 January 2027

I Proportion and proportion of a proportion

Definitions

In a non-empty finite set E, the proportion of a subset F is $p = \text{Card}(F)/\text{Card}(E)$; $0 \leq p \leq 1$. Expressed as a percentage, it is $100\,p$ %. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if p_1 is the proportion of F in E, p_2 that of G in F, the proportion of G in E is $p_1 p_2$.

Example 9.1

A lycée has 1500 students, 652 of whom are half-boarders. Calculate their proportion. In one class, 60 % of the students do sport; among them, 40 % are boys. What proportion of the class do the boys who do sport represent?

Solution:

Example 9.2

In a fictional school of 800 pupils, 35% do a club sport; among them, 45% do a team sport. Calculate the two numbers and the final proportion.

Solution:

Watch out. The proportion and the rate of change do not answer the same question: a proportion compares a part to a set, a rate compares a variation to an initial value.

II Absolute change and rate of change

Definition

For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

Example 9.3

A cheese is sold at 2 ,13 euros instead of 3 ,55 euros. Calculate the absolute change and the rate.

Solution:

Multiplier

$CM = V_F/V_I = 1 + t$, therefore $V_F = V_I(1 + t)$. To recover V_I, divide V_F by $1 + t$ (not zero).

Example 9.4

A hypothetical item costing 50000 Ar goes up by 12 %, then we consider separately a 12 % discount on the initial price. Give the final prices. A price after a 20% increase is 72000 Ar; find its initial.

Solution:

Fluency · 5 to 10 minutes

Expand $(x - 3)^2$; solve $1/x = 5$; give the mean of 8; 10; 12.

III Successive percentage changes

Property

Successive changes at rates t_1, t_2 have overall multiplier

$$CM_G = (1+t_1)(1+t_2), \qquad t_G = (1+t_1)(1+t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

Example 9.5

A manufacturer sold 5600 bicycles in 2022, i.e. 25 % more than in 2021. Its sales fall by 15 % in 2023. Determine the sales in 2021 and 2023, then the overall rate between those two years.

Solution:

Example 9.6

A price rises by 10 % then falls by 10 %. Does it return to the original price?

Solution:

Example 9.7

A rate rises from 20 % to 25 %. Explain the difference between the increase in percentage points and relative change.

Solution:

IV Reverse percentage change

Property

To return to the initial value after a change at rate $t > -1$, use the multiplier $1/(1+t)$.
The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

Example 9.8

What rate offsets a 25% increase? What rate offsets a 20% decrease?

Solution:

Example 9.9

A price is reduced by 30 %, then rises by 10 %. What additional percentage increase restores the original price?

Solution: